

An illustration of proportional hazards, additive hazards and accelerated failure time models

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Preliminaries

This document serves as the Supplementary Material of the paper “Alternatives to the Cox proportional hazards model: A review of additive hazards and accelerated failure time models”, (to be) published in Statistica Neerlandica. Section 5.2 of that paper contains a data example, of which this document shows the full R code.

We will be using the pbc data from the survival package, in a very much simplified setting. The aim of this illustration is to highlight differences in assumptions between different models, and to illustrate techniques that attempt to alleviate restrictions in the original models. We therefore restrict to only two covariates, hepato and edema, out of many in the pbc data that significantly influence survival. Of these two, edema violates the PH assumption.

```
options(warn = -1)
# Loading packages
suppressPackageStartupMessages({
  library(tictoc) # used for computation time
  library(survival)
  # Packages for additive hazards models, in addition to survival
  library(ahMLE)
```

```

library(timereg)
library(aftgee)
# Packages for AFT models, in addition to survival
library(lss2)
library(rms)
library(aftsem)
library(flexsurv)
library(eha)
})

# head(pbc) # look at help(pbc) for more information on the data
table(pbc$hepato, useNA = "always")

```

```

##
##      0      1 <NA>
## 152 160 106

```

One of the two variables of interest, `hepato`, is only used in the original 312 patients included in the RCT (see `help(pbc)` for context). For the purpose of this illustration we restrict to this subset. We are going to overwrite `pbc` (the original `pbc` data will remain available in the `survival` package).

```

pbc <- subset(pbc, !is.na(hepato))
# Looking at events
table(pbc$status) # 0=censored, 1=transplant, 2=dead

```

```

##
##      0      1      2
## 168 19 125

```

```

pbc$stat <- pbc$status
pbc$stat[pbc$stat == 2] <- 1 # so we use transplant-free survival
table(pbc$stat)

```

```

##
##      0      1
## 168 144

```

```

table(pbc$edema)

```

```

##
##      0 0.5      1
## 263 29 20

```

We are also going to overwrite `edema` with a binary version where the value 0.5 is set to 1 (so 0.5 and 1 are combined into one group). (Again, the original data will remain available in the `survival` package.)

```
pbcs$edema[pbcs$edema == 0.5] <- 1
```

In what follows we are going to construct model-based survival curves for each of the four combinations of hepato and edema. These are the numbers for these combinations.

```
table(pbc$hepato, pbc$edema)
```

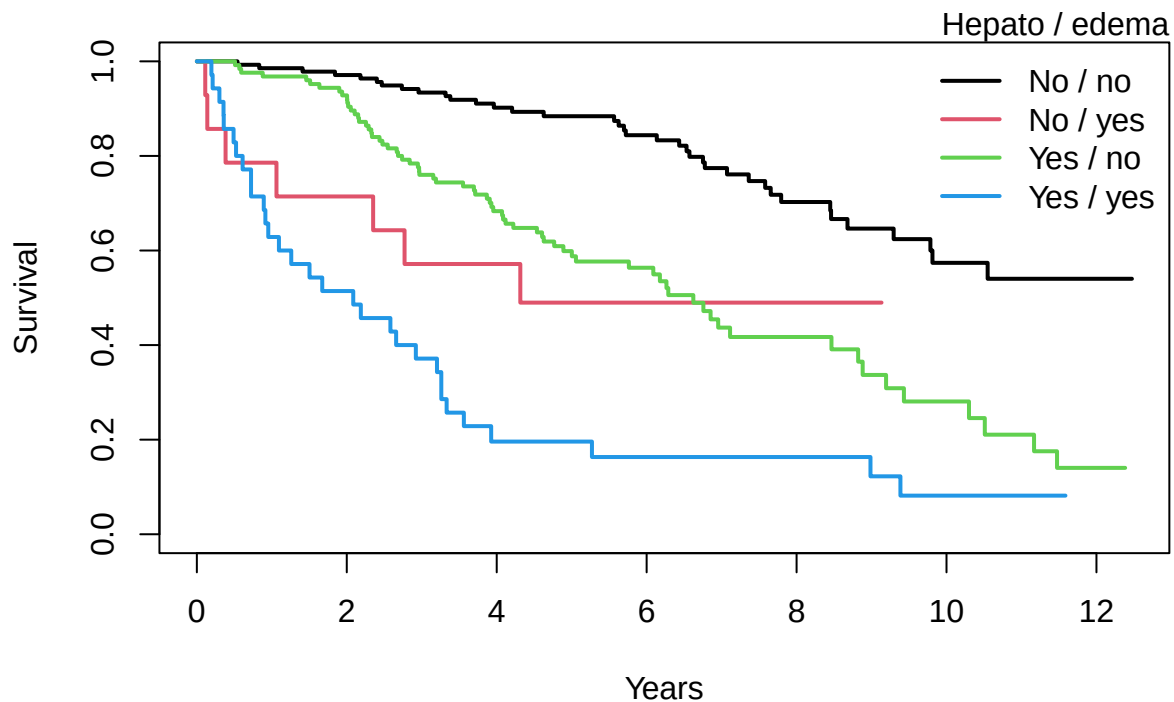
```
##
##      0    1
## 0 138  14
## 1 125  35

# We prefer to work with time in years
year <- 365.25
pbcs$yrs <- pbc$time / year
```

We start with separate Kaplan-Meier survival curves for the four different combinations no/yes of hepato and edema.

```
pbcs$he <- 2 * pbc$hepato + pbc$edema
pbcs$he <- factor(pbc$he, levels = 0:3,
                  labels = c("No / no", "No / yes", "Yes / no", "Yes / yes"))
sfhe <- survfit(Surv(yrs, stat) ~ he, data = pbc)
# pdf("Kaplan_Meiers.pdf")
plot(sfhe, lwd = 2, col = 1:4,
     xlab = "Years", ylab = "Survival")
legend("topright", c("No / no", "No / yes", "Yes / no", "Yes / yes"),
     lwd = 2, col = 1:4, bty = "n")
mtext("Hepato / edema", side = 3, adj = 1)
title(main = "Kaplan-Meier survival curves")
```

Kaplan-Meier survival curves



```
# dev.off()
```

Proportional hazards models

We start by fitting a Cox proportional hazards model with the two variables of interest.

```
c1 <- coxph(Surv(yrs, stat) ~ hepato + edema, data = pbc)
summary(c1) # both highly significant
```

```
## Call:
## coxph(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc)
##
##    n= 312, number of events= 144
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## hepato  1.0120    2.7511   0.1851  5.466 4.59e-08 ***
## edema   1.0591    2.8838   0.1936  5.471 4.48e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
```

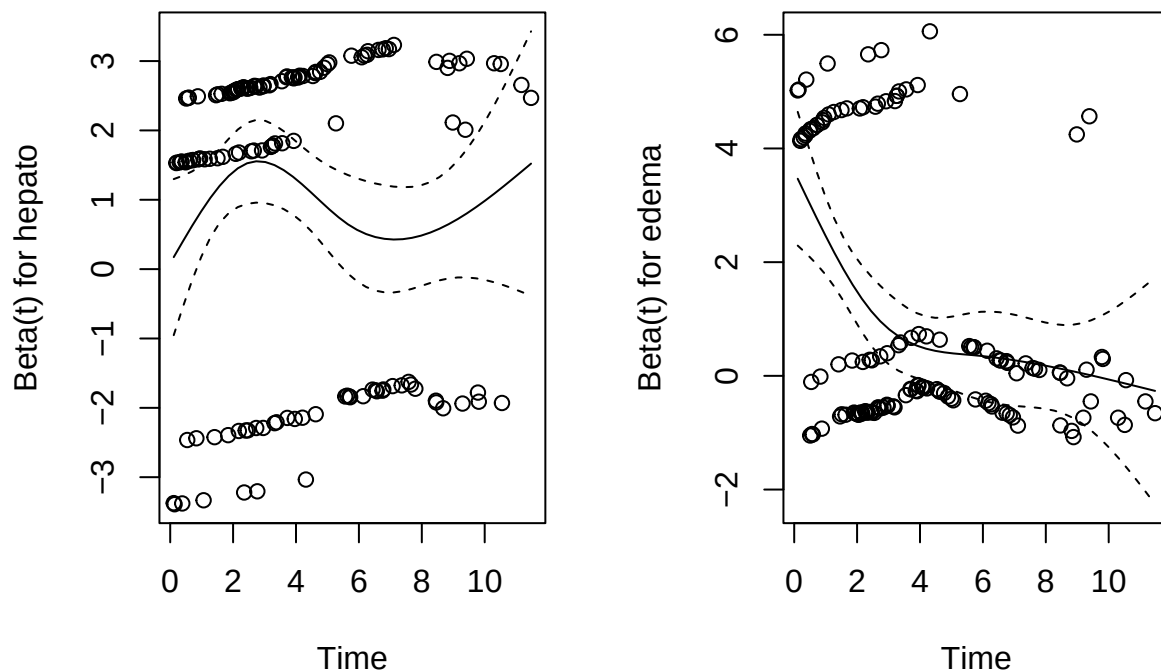
```
## hepato      2.751      0.3635      1.914      3.955
## edema       2.884      0.3468      1.973      4.214
##
## Concordance= 0.71 (se = 0.021 )
## Likelihood ratio test= 68.61 on 2 df, p=1e-15
## Wald test          = 72.7 on 2 df, p=<2e-16
## Score (logrank) test = 83.6 on 2 df, p=<2e-16
```

```
cz1 <- cox.zph(c1)
cz1 # edema violates PH
```

```
##          chisq df      p
## hepato   1.42  1    0.23
## edema   18.42  1 1.8e-05
## GLOBAL  18.43  2 1.0e-04
```

From the output of `cz1` we see that `edema` violates the proportional hazards (PH) assumption. The time-dependent behaviour $\beta(t)$ can be plotted, from which we can learn how the estimated regression coefficients varies with time. I am using `transform = "identity"` here to make sure that in the plot we see time linearly. The default is to plot $\beta(t)$ with time transformed, which makes it more difficult (in my view) to interpret $\beta(t)$.

```
par(mfrow = c(1, 2))
plot(cox.zph(c1, transform = "identity"))
```



```
par(mfrow = c(1, 1))
```

The $\beta(t)$ for **hepato** seems to go up and down somewhat, but one could imagine a straight line reasonably within the 95% pointwise confidence intervals, compatible with a time-constant β . The $\beta(t)$ for **edema** is quite different. The estimated β is huge, about 4 (this is the *log* hazard ratio!), and quickly decreases to 0 after about 5 years.

The easiest way to deal with the violation of the PH assumption for edema is to fit a stratified Cox model, where separate baseline hazards are estimated for edema no and edema yes, and a hazard ratio (HR) for hepato (assumed to be the same for edema no and edema yes).

```
# Stratified Cox model, stratified according to edema
c2 <- coxph(Surv(yrs, stat) ~ hepato + strata(edema), data = pbc)
summary(c2)
```

```
## Call:
## coxph(formula = Surv(yrs, stat) ~ hepato + strata(edema), data = pbc)
##
##      n= 312, number of events= 144
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## hepato  1.0538    2.8684    0.1867  5.643 1.67e-08 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##          exp(coef) exp(-coef) lower .95 upper .95
## hepato    2.868      0.3486    1.989    4.136
##
## Concordance= 0.638  (se = 0.023 )
## Likelihood ratio test= 34.56  on 1 df,   p=4e-09
## Wald test            = 31.84  on 1 df,   p=2e-08
## Score (logrank) test = 34.35  on 1 df,   p=5e-09
```

```
cox.zph(c2) # PH not violated for hepato
```

```
##          chisq df    p
## hepato 0.0215  1 0.88
## GLOBAL 0.0215  1 0.88
```

The PH assumption is not violated for hepato, so at least with respect to the PH assumption this seems to be a reasonable model. The model does not have an estimated hazard ratio / regression coefficient for edema. The reason is that the PH assumption for edema (which would give a HR) is replaced by two separate baseline hazards (so no HR). If a numeric description of the effect of edema is required, one would have to model $\beta(t)$ by replacing the stratification by an interaction of edema with a pre-specified function of time. The plot of `cox.zph` above could inform which function to use. Judging by this plot, something like $f(t) = \log(t + 1)$ could work well.

We will continue to show survival curves for four patients, namely for all four combinations of hepato 0/1, and edema 0/1.

```
# According to Cox model, and stratified Cox model
# First Cox model
nd00 <- data.frame(hepato = 0, edema = 0)
nd01 <- data.frame(hepato = 0, edema = 1)
nd10 <- data.frame(hepato = 1, edema = 0)
nd11 <- data.frame(hepato = 1, edema = 1)

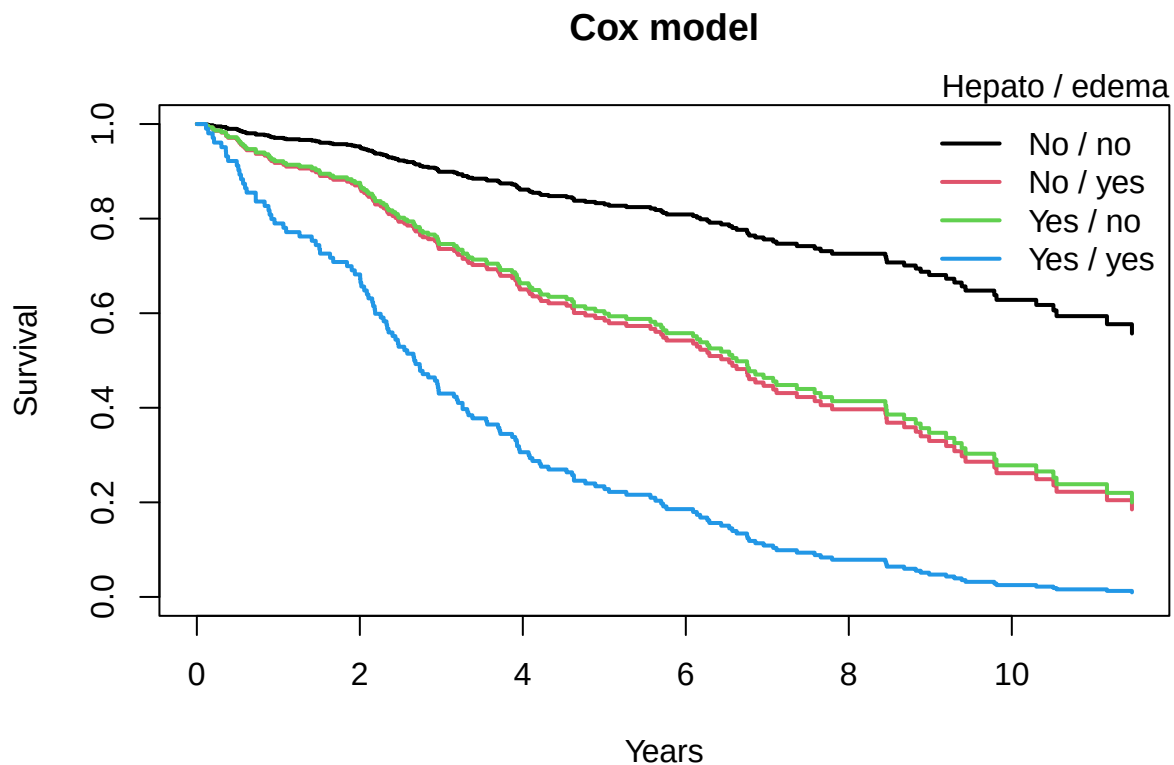
sf00 <- survfit(c1, newdata = nd00)
sf00 <- data.frame(time = sf00$time, surv = sf00$surv)
sf00 <- rbind(data.frame(time = 0, surv = 1), sf00)
sf00 <- subset(sf00, !duplicated(surv))
sf01 <- survfit(c1, newdata = nd01)
sf01 <- data.frame(time = sf01$time, surv = sf01$surv)
sf01 <- rbind(data.frame(time = 0, surv = 1), sf01)
sf01 <- subset(sf01, !duplicated(surv))
sf10 <- survfit(c1, newdata = nd10)
sf10 <- data.frame(time = sf10$time, surv = sf10$surv)
sf10 <- rbind(data.frame(time = 0, surv = 1), sf10)
sf10 <- subset(sf10, !duplicated(surv))
```

```

sf11 <- survfit(c1, newdata = nd11)
sf11 <- data.frame(time = sf11$time, surv = sf11$surv)
sf11 <- rbind(data.frame(time = 0, surv = 1), sf11)
sf11 <- subset(sf11, !duplicated(surv))

# pdf("Cox_model.pdf")
plot(sf00$time, sf00$surv, type = "s", ylim = c(0, 1), lwd = 2,
      xlab = "Years", ylab = "Survival")
lines(sf01$time, sf01$surv, type = "s", col = 2, lwd = 2)
lines(sf10$time, sf10$surv, type = "s", col = 3, lwd = 2)
lines(sf11$time, sf11$surv, type = "s", col = 4, lwd = 2)
legend("topright", c("No / no", "No / yes", "Yes / no", "Yes / yes"),
      lwd = 2, col = 1:4, bty = "n")
mtext("Hepato / edema", side = 3, adj = 1)
title(main = "Cox model")

```



```

# dev.off()

sf00_cox <- sf00
sf01_cox <- sf01
sf10_cox <- sf10

```



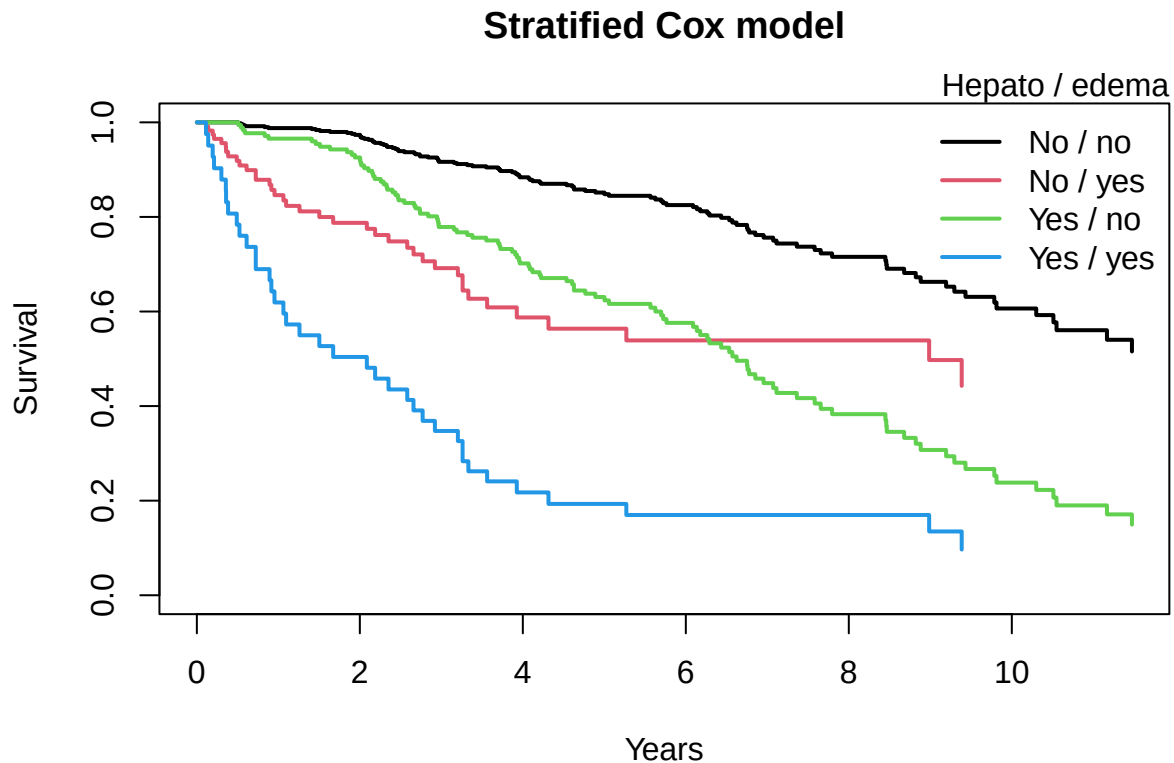
```

sf11_cox <- sf11

# Then stratified Cox model
sf00 <- survfit(c2, newdata = nd00)
sf00 <- data.frame(time = sf00$time, surv = sf00$surv)
sf00 <- rbind(data.frame(time = 0, surv = 1), sf00)
sf00 <- subset(sf00, !duplicated(surv))
sf01 <- survfit(c2, newdata = nd01)
sf01 <- data.frame(time = sf01$time, surv = sf01$surv)
sf01 <- rbind(data.frame(time = 0, surv = 1), sf01)
sf01 <- subset(sf01, !duplicated(surv))
sf10 <- survfit(c2, newdata = nd10)
sf10 <- data.frame(time = sf10$time, surv = sf10$surv)
sf10 <- rbind(data.frame(time = 0, surv = 1), sf10)
sf10 <- subset(sf10, !duplicated(surv))
sf11 <- survfit(c2, newdata = nd11)
sf11 <- data.frame(time = sf11$time, surv = sf11$surv)
sf11 <- rbind(data.frame(time = 0, surv = 1), sf11)
sf11 <- subset(sf11, !duplicated(surv))

# pdf("Stratified_Cox_model.pdf")
plot(sf00$time, sf00$surv, type = "s", ylim = c(0, 1), lwd = 2,
      xlab = "Years", ylab = "Survival")
lines(sf01$time, sf01$surv, type = "s", col = 2, lwd = 2)
lines(sf10$time, sf10$surv, type = "s", col = 3, lwd = 2)
lines(sf11$time, sf11$surv, type = "s", col = 4, lwd = 2)
legend("topright", c("No / no", "No / yes", "Yes / no", "Yes / yes"),
      lwd = 2, col = 1:4, bty = "n")
mtext("Hepato / edema", side = 3, adj = 1)
title(main = "Stratified Cox model")

```



```
# dev.off()
```

Additive hazards models

We continue to fit additive hazards models. The non-parametric OLS has been implemented for instance in the survival package, function `aareg()`, and in the timereg package, using `aalen()`. We will first illustrate `aareg()` from survival.

```
a0 <- aareg(Surv(yrs, stat) ~ hepato + edema, data = pbc)
head(a0$coef) # Estimated beta's
```

```
##               Intercept      hepato      edema
## 0.112251882272416 0.004491233 -0.009449554 0.02266661
## 0.139630390143737 0.004616610 -0.009713348 0.02329937
## 0.194387405886379 -0.001628293 0.003425929 0.02035367
## 0.210814510609172 -0.001665179 0.003503537 0.02081474
## 0.301163586584531 -0.001703775 0.003584742 0.02129719
## 0.355920602327173 -0.001744202 0.003669802 0.02180253
```

```
head(apply(a0$coef, 2, cumsum)) # Estimated cumulative beta's
```

```
##               Intercept      hepato      edema
```

```
## 0.112251882272416 0.004491233 -0.009449554 0.02266661
## 0.139630390143737 0.009107843 -0.019162903 0.04596598
## 0.194387405886379 0.007479550 -0.015736974 0.06631965
## 0.210814510609172 0.005814371 -0.012233437 0.08713439
## 0.301163586584531 0.004110596 -0.008648694 0.10843157
## 0.355920602327173 0.002366394 -0.004978893 0.13023410
```

```
# plot(a0)
```

The function `aalen()` from the `timereg` package gives exactly the same estimates as `aareg()` from `survival`, but in addition useful tests for $\beta(t) = 0$ and for $\beta(t)$ being time-constant are also provided (for the latter null hypothesis both a Kolmogorov-Smirnov and a Cramer-von Mises test).

```
a1 <- aalen(Surv(yrs, stat) ~ hepato + edema, data = pbc)
head(a1$cum) # same as the cumsum of a0$coef
```

```
##           time (Intercept)      hepato      edema
## [1,] 0.0000000 0.000000000 0.000000000 0.00000000
## [2,] 0.1122519 0.004491233 -0.009449554 0.02266661
## [3,] 0.1396304 0.009107843 -0.019162903 0.04596598
## [4,] 0.1943874 0.007479550 -0.015736974 0.06631965
## [5,] 0.2108145 0.005814371 -0.012233437 0.08713439
## [6,] 0.3011636 0.004110596 -0.008648694 0.10843157
```

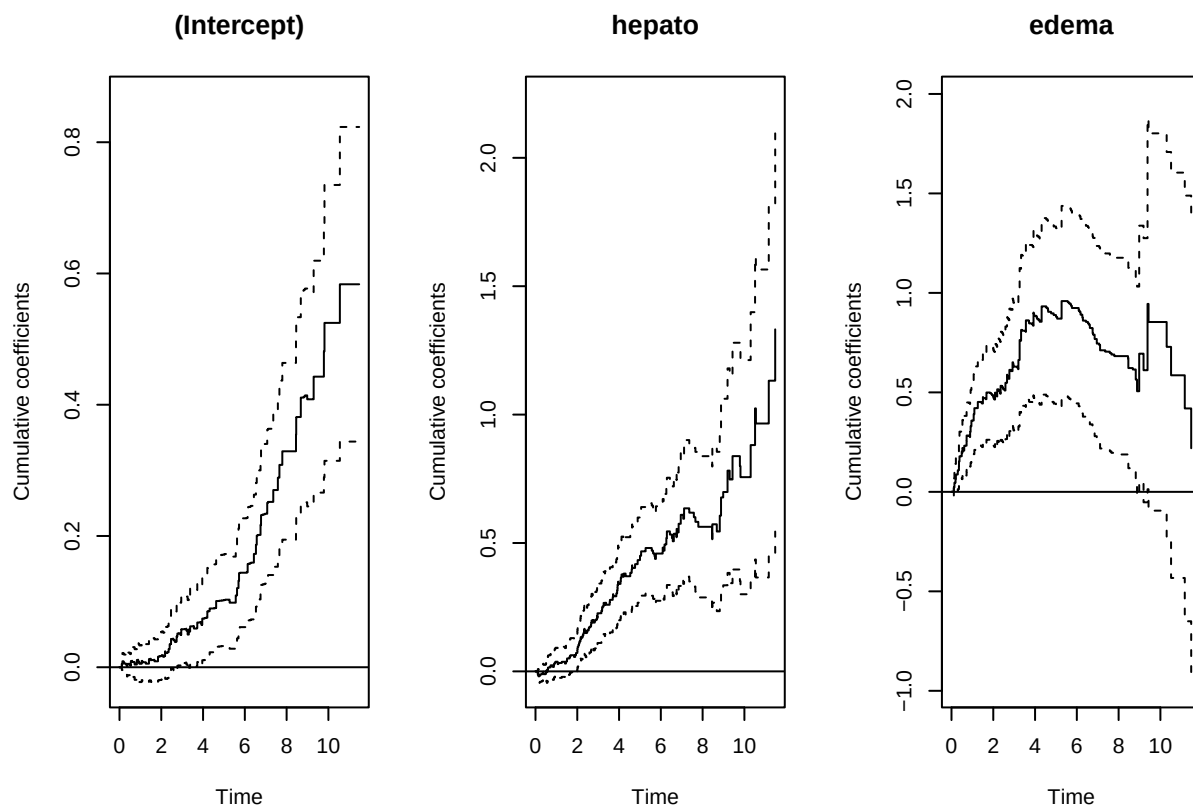
```
summary(a1)
```

```
## Additive Aalen Model
##
## Test for nonparametric terms
##
## Test for non-significant effects
##           Supremum-test of significance p-value H_0: B(t)=0
## (Intercept)                    5.01                      0
## hepato                        5.33                      0
## edema                         4.43                      0
##
## Test for time invariant effects
##           Kolmogorov-Smirnov test
## (Intercept)                    0.174
## hepato                        0.468
## edema                         0.858
##           p-value H_0:constant effect
## (Intercept)                    0.043
## hepato                        0.182
## edema                         0.028
##           Cramer von Mises test
```

```
## (Intercept)                0.109
## hepato                     0.601
## edema                      3.890
##           p-value H_0:constant effect
## (Intercept)                0.027
## hepato                     0.197
## edema                      0.009
##
##
## Call:
## aalen(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc)
```

Both hepato and edema have a significant effect on survival (null hypothesis of $\beta(t) = 0$ is rejected). The null hypothesis of $\beta(t)$ being constant is rejected for edema, not for hepato (note that this assumption is not the PH assumption). A plot function for the result is provided.

```
# pdf("Ah_Bt.pdf", width = 8, height = 5)
par(mfrow = c(1, 3))
plot(a1, pointwise.ci = 2)
```



```
# dev.off()
```

We now derive the survival curves for the same four patients.

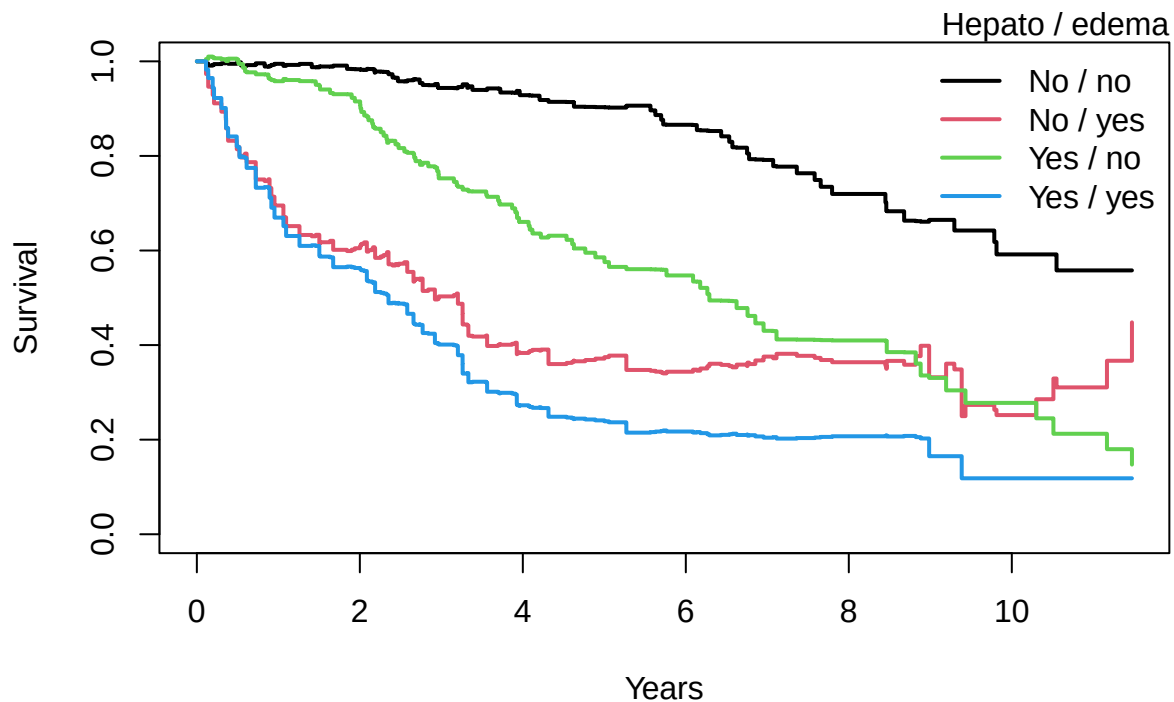
```
ah1 <- a1$cum
ah1 <- as.data.frame(ah1)
names(ah1) <- dimnames(ah1)[[2]]
names(ah1)[2] <- "intercept"
head(ah1)

##      time  intercept      hepato      edema
## 1 0.0000000 0.000000000 0.000000000 0.00000000
## 2 0.1122519 0.004491233 -0.009449554 0.02266661
## 3 0.1396304 0.009107843 -0.019162903 0.04596598
## 4 0.1943874 0.007479550 -0.015736974 0.06631965
## 5 0.2108145 0.005814371 -0.012233437 0.08713439
## 6 0.3011636 0.004110596 -0.008648694 0.10843157

b00 <- data.frame(time = ah1$time, Haz = ah1$intercept)
b00$surv <- exp(-b00$Haz)
b01 <- data.frame(time = ah1$time, Haz = ah1$intercept + ah1$edema)
b01$surv <- exp(-b01$Haz)
b10 <- data.frame(time = ah1$time, Haz = ah1$intercept + ah1$hepato)
b10$surv <- exp(-b10$Haz)
b11 <- data.frame(time = ah1$time, Haz = ah1$intercept + ah1$hepato + ah1$edema)
b11$surv <- exp(-b11$Haz)

# pdf("Ah_nonpar_survival.pdf")
par(mfrow = c(1, 1))
plot(b00$time, b00$surv, type = "s", ylim = c(0, 1), lwd = 2,
     xlab = "Years", ylab = "Survival")
lines(b01$time, b01$surv, type = "s", col = 2, lwd = 2)
lines(b10$time, b10$surv, type = "s", col = 3, lwd = 2)
lines(b11$time, b11$surv, type = "s", col = 4, lwd = 2)
legend("topright", c("No / no", "No / yes", "Yes / no", "Yes / yes"),
     lwd = 2, col = 1:4, bty = "n")
mtext("Hepato / edema", side = 3, adj = 1)
title(main = "Additive hazards model (OLS)")
```

Additive hazards model (OLS)



```
# dev.off()
```

```
b00_OLS <- b00
b01_OLS <- b01
b10_OLS <- b10
b11_OLS <- b11
```

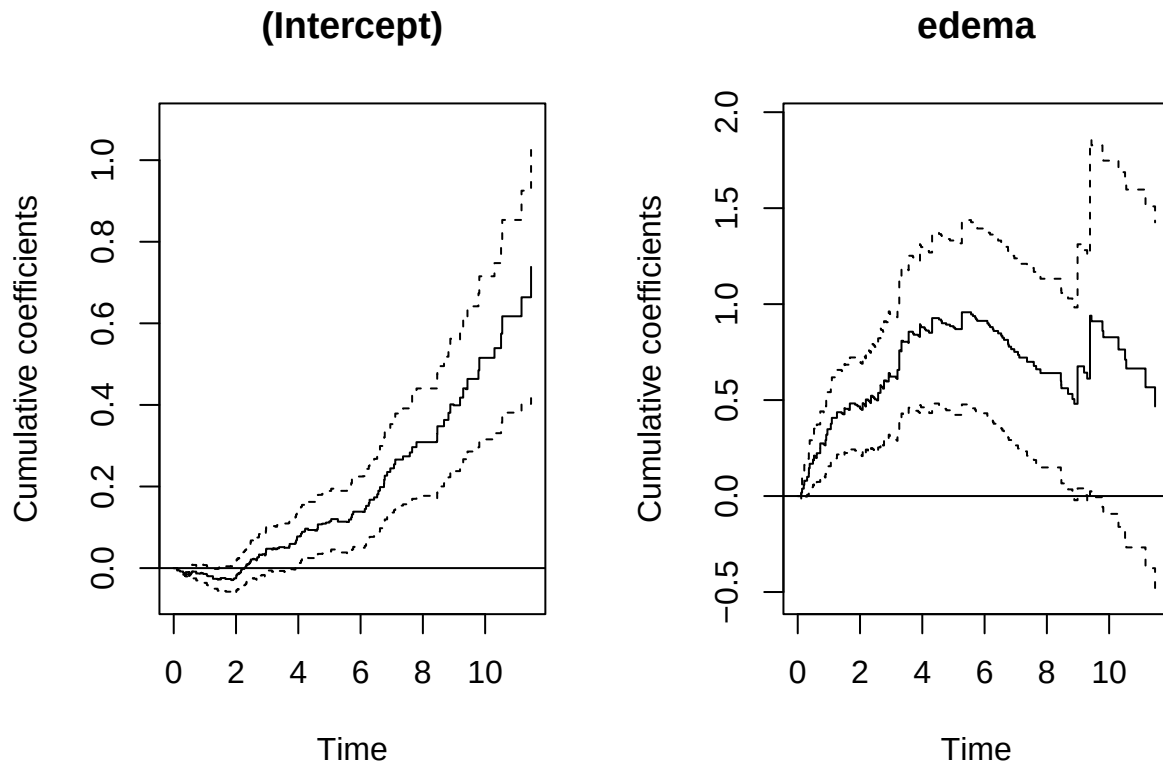
Note the non-monotone behavior of in particular the hepato no, edema yes curve.

Since the effect of hepato seemed to be time-constant, we also fit a model with a constant effect of hepato.

```
a2 <- aalen(Surv(yrs, stat) ~ const(hepato) + edema, data = pbc)
summary(a2)
```

```
## Additive Aalen Model
##
## Test for nonparametric terms
##
## Test for non-significant effects
##
##      Supremum-test of significance p-value H_0: B(t)=0
## (Intercept)          5.29                0
## edema                4.42                0
```

```
##
## Test for time invariant effects
##           Kolmogorov-Smirnov test
## (Intercept)           0.254
## edema                 0.752
##           p-value H_0:constant effect
## (Intercept)           0.044
## edema                 0.007
##           Cramer von Mises test
## (Intercept)           0.339
## edema                 2.500
##           p-value H_0:constant effect
## (Intercept)           0.008
## edema                 0.003
##
## Parametric terms :
##           Coef.      SE Robust SE      z      P-val lower2.5%
## const(hepato) 0.0828 0.0148      0.0147 5.62 1.89e-08      0.0538
##           upper97.5%
## const(hepato)      0.112
##
## Call:
## aalen(formula = Surv(yrs, stat) ~ const(hepato) + edema, data = pbc)
par(mfrow = c(1, 2))
plot(a2, pointwise.ci = 2)
```



The estimated cumulative curves for intercept (cumulative baseline hazard) and **edema** have not fundamentally changed, but note that the cumulative baseline hazard estimate is negative until just past $t = 2$. Also for this model we derive the survival curves for the same four patients.

```
ah2 <- a2$cum
ah2 <- as.data.frame(ah2)
names(ah2) <- dimnames(ah2)[[2]]
names(ah2)[2] <- "intercept"
head(ah2)
```

```
##      time    intercept    edema
## 1 0.000000  0.000000000 0.0000000
## 2 0.1122519 -0.004418756 0.01818616
## 3 0.1396304 -0.005496501 0.03844380
## 4 0.1943874 -0.007651992 0.05849864
## 5 0.2108145 -0.008298639 0.07987880
## 6 0.3011636 -0.011855198 0.10017005
```

```
b00 <- data.frame(time = ah2$time, Haz = ah2$intercept)
b00$surv <- exp(-b00$Haz)
b01 <- data.frame(time = ah2$time, Haz = ah2$intercept + ah2$edema)
b01$surv <- exp(-b01$Haz)
```

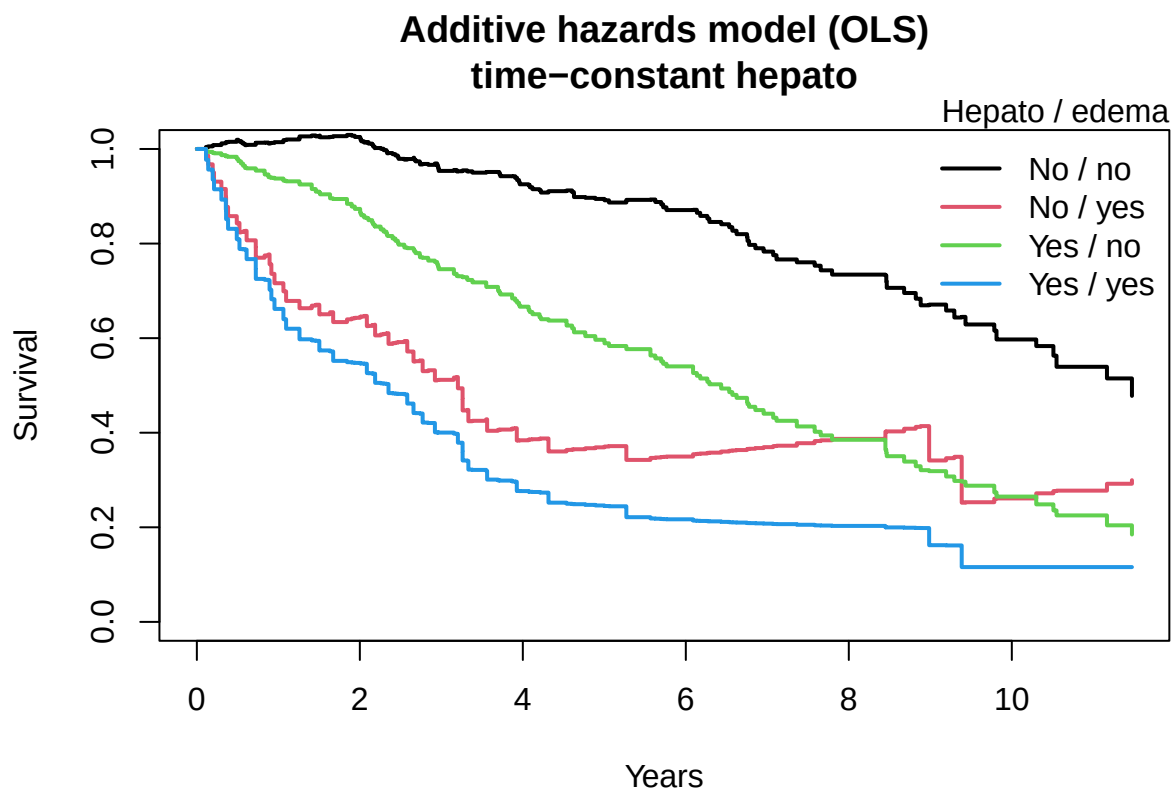


```

b10 <- data.frame(time = ah2$time, Haz = ah2$intercept + ah2$time * c(a2$gamma))
b10$surv <- exp(-b10$Haz)
b11 <- data.frame(time = ah2$time, Haz = ah2$intercept + ah2$time * c(a2$gamma) + ah2$edema)
b11$surv <- exp(-b11$Haz)

# pdf("Ah_McKS_survival.pdf")
par(mfrow = c(1, 1))
plot(b00$time, b00$surv, type = "s", ylim = c(0, 1), lwd = 2,
     xlab = "Years", ylab = "Survival")
lines(b01$time, b01$surv, type = "s", col = 2, lwd = 2)
lines(b10$time, b10$surv, type = "s", col = 3, lwd = 2)
lines(b11$time, b11$surv, type = "s", col = 4, lwd = 2)
legend("topright", c("No / no", "No / yes", "Yes / no", "Yes / yes"),
     lwd = 2, col = 1:4, bty = "n")
mtext("Hepato / edema", side = 3, adj = 1)
title(main = "Additive hazards model (OLS)\ntime-constant hepato")

```



```
# dev.off()
```

We can now also see the model-based survival curve for no hepato, no edema being larger than one (consistent with the cumulative baseline hazard being negative).

```

b00_OLS_tc <- b00
b01_OLS_tc <- b01
b10_OLS_tc <- b10
b11_OLS_tc <- b11

```

Finally we consider the MLE estimates of Lu et al. (2023).

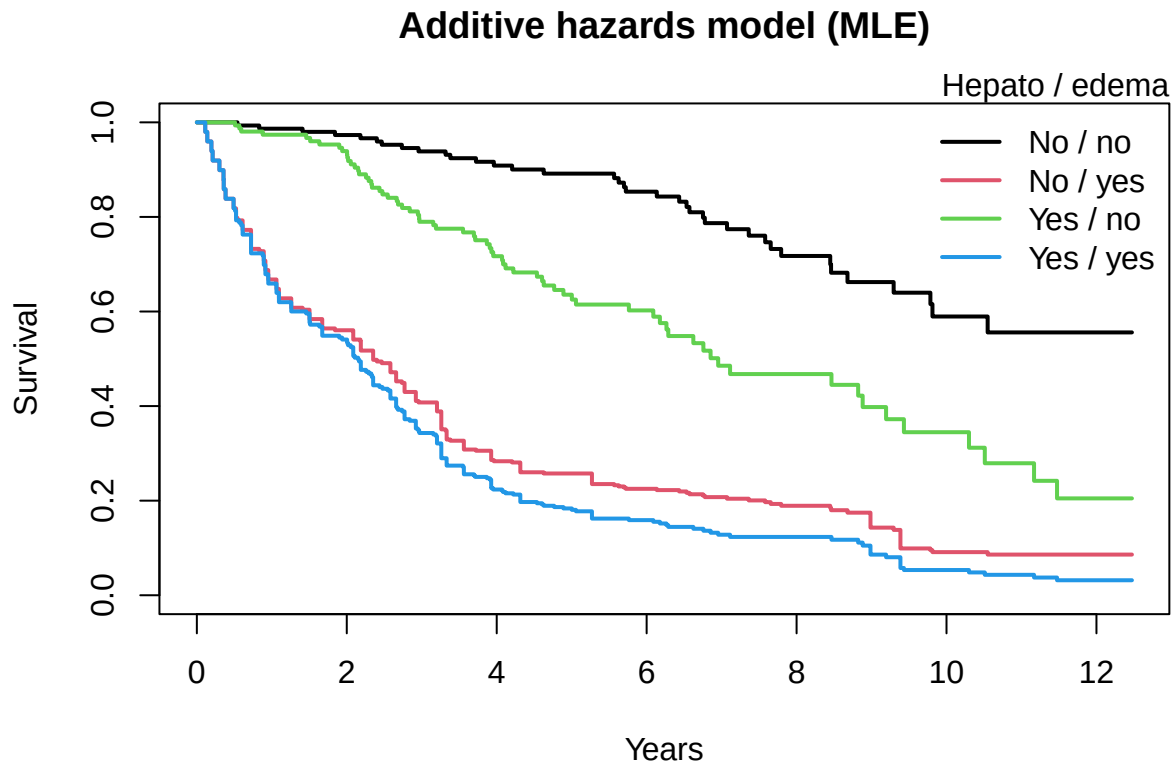
```

a3 <- ah(Surv(yrs, stat) ~ hepato + edema, data = pbc)
a3 <- a3$cumbeta

# Survival curves for the same four patients
b00 <- data.frame(time = a3$time, Haz = a3$intercept)
b00$surv <- exp(-b00$Haz)
b01 <- data.frame(time = a3$time, Haz = a3$intercept + a3$edema)
b01$surv <- exp(-b01$Haz)
b10 <- data.frame(time = a3$time, Haz = a3$intercept + a3$hepato)
b10$surv <- exp(-b10$Haz)
b11 <- data.frame(time = a3$time, Haz = a3$intercept + a3$hepato + a3$edema)
b11$surv <- exp(-b11$Haz)

# pdf("Ah_MLE_survival.pdf")
par(mfrow = c(1, 1))
plot(b00$time, b00$surv, type = "s", ylim = c(0, 1), lwd = 2,
     xlab = "Years", ylab = "Survival")
lines(b01$time, b01$surv, type = "s", col = 2, lwd = 2)
lines(b10$time, b10$surv, type = "s", col = 3, lwd = 2)
lines(b11$time, b11$surv, type = "s", col = 4, lwd = 2)
legend("topright", c("No / no", "No / yes", "Yes / no", "Yes / yes"),
     lwd = 2, col = 1:4, bty = "n")
mtext("Hepato / edema", side = 3, adj = 1)
title(main = "Additive hazards model (MLE)")

```



```
# dev.off()
```

Note that all mode-based survival curves are monotone, but especially the hepato no, edema yes curve is quite different from the previous additive hazard curves.

```
b00_MLE <- b00
b01_MLE <- b01
b10_MLE <- b10
b11_MLE <- b11
```

AFT models

Finally we consider AFT models, first parametric, then semi-parametric AFT models.

Parametric AFT models

We start with parametric AFT models, first `survreg()` from the `survival` package.

```
tic("survreg from {survival}")
ls.sur.weib <- survreg(Surv(yrs, stat) ~ hepato + edema, data = pbc)
toc()
```

```
## survreg from {survival}: 0.01 sec elapsed
```

```
summary(ls.sur.weib)
```

```
##
## Call:
## survreg(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc)
##               Value Std. Error      z      p
## (Intercept)  2.9566      0.1387 21.32 < 2e-16
## hepato      -0.8141      0.1529 -5.32 1.0e-07
## edema       -0.8674      0.1613 -5.38 7.5e-08
## Log(scale)  -0.2260      0.0695 -3.25 0.0011
##
## Scale= 0.798
##
## Weibull distribution
## Loglik(model)= -463.4   Loglik(intercept only)= -498.6
##  Chisq= 70.44 on 2 degrees of freedom, p= 5.1e-16
## Number of Newton-Raphson Iterations: 5
## n= 312
```

It is instructive to see what happens if we use the original time scale, days, rather than years. From the results we see that the estimated regression coefficients for hepato and edema remain the same, but the intercept has changed.

```
tic("survreg from {survival}, time in days")
ls.sur.weib2 <- survreg(Surv(time, stat) ~ hepato + edema, data = pbc)
toc()
```

```
## survreg from {survival}, time in days: 0 sec elapsed
```

```
summary(ls.sur.weib2)
```

```
##
## Call:
## survreg(formula = Surv(time, stat) ~ hepato + edema, data = pbc)
##               Value Std. Error      z      p
## (Intercept)  8.8572      0.1387 63.86 < 2e-16
## hepato      -0.8141      0.1529 -5.32 1.0e-07
## edema       -0.8674      0.1613 -5.38 7.5e-08
## Log(scale)  -0.2260      0.0695 -3.25 0.0011
##
## Scale= 0.798
##
## Weibull distribution
## Loglik(model)= -1313   Loglik(intercept only)= -1348.3
##  Chisq= 70.44 on 2 degrees of freedom, p= 5.1e-16
```

```
## Number of Newton-Raphson Iterations: 5
## n= 312
```

Many other parametric distributions can be used in `survreg()`, like for instance the logistic distributions.

```
tic("survreg with logistic distribution")
ls.sur.logistic <- survreg(Surv(yrs, stat) ~ hepato + edema, data = pbc,
                           dist = "logistic")
toc()
```

```
## survreg with logistic distribution: 0.02 sec elapsed
```

```
summary(ls.sur.logistic)
```

```
##
## Call:
## survreg(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc,
##         dist = "logistic")
##              Value Std. Error      z      p
## (Intercept) 10.3073     0.4935 20.89 < 2e-16
## hepato      -3.7487     0.5887 -6.37 1.9e-10
## edema       -4.1063     0.6748 -6.09 1.2e-09
## Log(scale)   0.8716     0.0685 12.73 < 2e-16
##
## Scale= 2.39
##
## Logistic distribution
## Loglik(model)= -497.5   Loglik(intercept only)= -539.8
##  Chisq= 84.64 on 2 degrees of freedom, p= 4.2e-19
## Number of Newton-Raphson Iterations: 4
## n= 312
```

About the parametrization of the Weibull distribution. The function `flexsurvreg()` in the `flexsurv` package fits AFT Weibull models with a parametrization more in line with that used in the paper (and models with other parametric error distributions). So let's fit the Weibull AFT model using `flexsurvreg()`, and also AFT models with a range of other distributions, including the generalized gamma distribution.

```
fitw <- flexsurvreg(Surv(yrs, stat) ~ hepato + edema, data = pbc, dist="weibull")
fitl <- flexsurvreg(Surv(yrs, stat) ~ hepato + edema, data = pbc, dist="llogis")
fitg <- flexsurvreg(Surv(yrs, stat) ~ hepato + edema, data = pbc, dist="gengamma")
fitn <- flexsurvreg(Surv(yrs, stat) ~ hepato + edema, data = pbc, dist="lnorm")
fitf <- flexsurvreg(Surv(yrs, stat) ~ hepato + edema, data = pbc, dist="genf")
fitg <- flexsurvreg(Surv(yrs, stat) ~ hepato + edema, data = pbc, dist="gompertz")
fitw
```

```
## Call:
```

```
## flexsurvreg(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc,
##     dist = "weibull")
##
## Estimates:
##      data mean  est      L95%      U95%      se      exp(est)
## shape      NA    1.2536   1.0940   1.4365   0.0871      NA
## scale      NA   19.2331  14.6553  25.2409   2.6675      NA
## hepato    0.5128   -0.8141  -1.1137  -0.5144   0.1529   0.4431
## edema     0.1571   -0.8674  -1.1835  -0.5514   0.1613   0.4200
##      L95%      U95%
## shape      NA      NA
## scale      NA      NA
## hepato    0.3283   0.5978
## edema     0.3062   0.5762
##
## N = 312,  Events: 144,  Censored: 168
## Total time at risk: 1713.854
## Log-likelihood = -463.3634, df = 4
## AIC = 934.7267
```

```
fitl
```

```
## Call:
## flexsurvreg(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc,
##     dist = "llogis")
##
## Estimates:
##      data mean  est      L95%      U95%      se      exp(est)  L95%
## shape      NA    1.637   1.425   1.880   0.116      NA      NA
## scale      NA   14.396  11.309  18.325   1.773      NA      NA
## hepato    0.513   -0.815  -1.102  -0.528   0.146   0.443   0.332
## edema     0.157   -1.265  -1.634  -0.896   0.188   0.282   0.195
##      U95%
## shape      NA
## scale      NA
## hepato    0.590
## edema     0.408
##
## N = 312,  Events: 144,  Censored: 168
## Total time at risk: 1713.854
## Log-likelihood = -457.5594, df = 4
## AIC = 923.1187
```

```
fitg
```

```
## Call:
```

```
## flexsurvreg(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc,
##     dist = "gompertz")
##
## Estimates:
##      data mean  est      L95%    U95%    se      exp(est)  L95%
## shape      NA    0.0871 0.0306 0.1435 0.0288      NA      NA
## rate       NA    0.0283 0.0191 0.0420 0.0057      NA      NA
## hepato    0.5128    1.0209 0.6587 1.3831 0.1848  2.7756  1.9322
## edema    0.1571    1.0633 0.6863 1.4404 0.1924  2.8960  1.9863
##          U95%
## shape      NA
## rate       NA
## hepato    3.9872
## edema    4.2223
##
## N = 312,  Events: 144,  Censored: 168
## Total time at risk: 1713.854
## Log-likelihood = -463.8338, df = 4
## AIC = 935.6676
```

```
fitn
```

```
## Call:
## flexsurvreg(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc,
##     dist = "lnorm")
##
## Estimates:
##      data mean  est      L95%    U95%    se      exp(est)
## meanlog      NA    2.7158  2.4557  2.9759 0.1327      NA
## sdlog        NA    1.1186  0.9898  1.2642 0.0698      NA
## hepato    0.5128   -0.7901 -1.0936 -0.4865 0.1549  0.4538
## edema    0.1571   -1.4046 -1.7724 -1.0368 0.1877  0.2455
##          L95%    U95%
## meanlog      NA      NA
## sdlog        NA      NA
## hepato    0.3350  0.6147
## edema    0.1699  0.3546
##
## N = 312,  Events: 144,  Censored: 168
## Total time at risk: 1713.854
## Log-likelihood = -459.3138, df = 4
## AIC = 926.6276
```

```
fitf
```

```
## Call:
## flexsurvreg(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc,
```

```
##      dist = "genf")
##
## Estimates:
##      data mean  est      L95%    U95%    se      exp(est)  L95%
## mu           NA    2.544    2.141    2.947    0.206      NA        NA
## sigma        NA    0.821    0.558    1.209    0.162      NA        NA
## Q            NA   -0.337   -1.257    0.583    0.469      NA        NA
## P            NA    1.588    0.330    7.645    1.273      NA        NA
## hepato       0.513   -0.800   -1.086   -0.514    0.146    0.449    0.338
## edema        0.157   -1.354   -1.820   -0.888    0.238    0.258    0.162
##      U95%
## mu           NA
## sigma        NA
## Q            NA
## P            NA
## hepato       0.598
## edema        0.412
##
## N = 312,  Events: 144,  Censored: 168
## Total time at risk: 1713.854
## Log-likelihood = -457.254, df = 6
## AIC = 926.508
```

```
fitg
```

```
## Call:
## flexsurvreg(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc,
##      dist = "gompertz")
##
## Estimates:
##      data mean  est      L95%    U95%    se      exp(est)  L95%
## shape        NA    0.0871  0.0306  0.1435  0.0288      NA        NA
## rate         NA    0.0283  0.0191  0.0420  0.0057      NA        NA
## hepato       0.5128    1.0209  0.6587  1.3831  0.1848    2.7756    1.9322
## edema        0.1571    1.0633  0.6863  1.4404  0.1924    2.8960    1.9863
##      U95%
## shape        NA
## rate         NA
## hepato       3.9872
## edema        4.2223
##
## N = 312,  Events: 144,  Censored: 168
## Total time at risk: 1713.854
## Log-likelihood = -463.8338, df = 4
## AIC = 935.6676
```


Package eha has the function `aftreg()` that fits parametric AFT models. Default is the Weibull distributions, but also loglogistic, lognormal and Gompertz distributions are supported. The same results are obtained as with other packages, at least with respect to the regression coefficients. It is unclear where for instance the shape and scale of the Weibull model can be found.

```
aftregw <- aftreg(Surv(yrs, stat) ~ hepato + edema, data = pbc)
summary(aftregw)
```

```
## Covariate           W.mean      Coef Time-Accn  se(Coef)    LR p
## hepato              0.432      0.814     2.257     0.153    0.0000
## edema               0.094      0.868     2.381     0.161    0.0000
##
## Events              144
## Total time at risk  1713.9
## Max. log. likelihood -463.36
## LR test statistic    70.44
## Degrees of freedom   2
## Overall p-value      5.55112e-16
```

```
aftregll <- aftreg(Surv(yrs, stat) ~ hepato + edema, data = pbc,
                    dist = "loglogistic")
summary(aftregll)
```

```
## Covariate           W.mean      Coef Time-Accn  se(Coef)    LR p
## hepato              0.432      0.815     2.259     0.146    0.0000
## edema               0.094      1.265     3.543     0.188    0.0000
##
## Events              144
## Total time at risk  1713.9
## Max. log. likelihood -457.56
## LR test statistic    85.52
## Degrees of freedom   2
## Overall p-value      0
```

The function `psm()` from the rms package also fits AFT models. From this it also seems to be easier to get predicted survival functions. The following code provides model-based survival curves for the four groups.

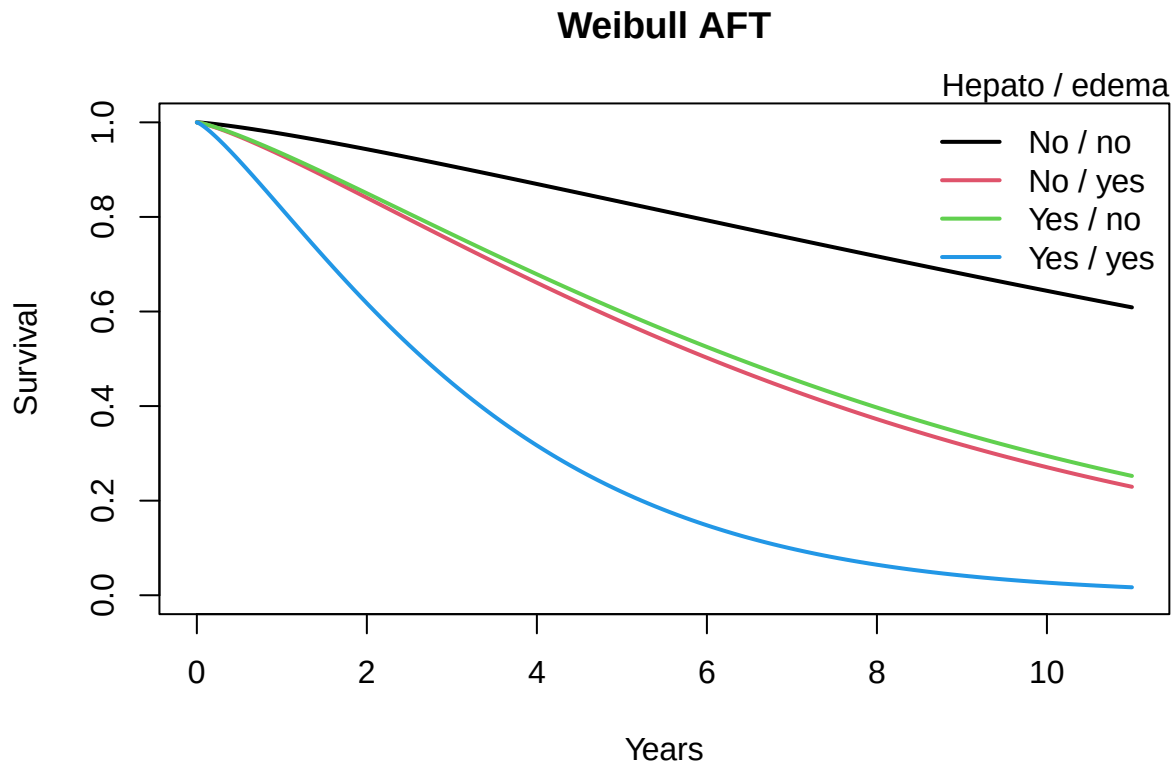
```
mySweibull <- function(x, rate, shape)
{
  return(exp( - rate * x^shape))
}
psm.weib <- psm(Surv(yrs, stat) ~ hepato + edema, data = pbc)
psm.weib
```

```
## Parametric Survival Model: Weibull Distribution
```

```
##
## psm(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc)
##
##               Model Likelihood      Discrimination
##               Ratio Test              Indexes
## Obs          312    LR chi2         70.44    R2          0.211
## Events       144    d.f.             2      R2(2,312)0.197
## sigma 0.7977    Pr(> chi2) <0.0001    R2(2,144)0.378
##                                     Dxy          0.421
##
##           Coef      S.E.   Wald Z Pr(>|Z|)
## (Intercept)  2.9566 0.1387 21.32  <0.0001
## hepato       -0.8141 0.1529 -5.32  <0.0001
## edema        -0.8674 0.1613 -5.38  <0.0001
## Log(scale)  -0.2260 0.0695 -3.25  0.0011

sig <- psm.weib$scale
lp <- predict(psm.weib, newdata = nd00, type = "lp")
tseq <- seq(0, 11, by = 0.01)
p00 <- mySweibull(tseq, shape = 1 / sig, rate = exp(-lp / sig))
lp <- predict(psm.weib, newdata = nd01, type = "lp")
p01 <- mySweibull(tseq, shape = 1 / sig, rate = exp(-lp / sig))
lp <- predict(psm.weib, newdata = nd10, type = "lp")
p10 <- mySweibull(tseq, shape = 1 / sig, rate = exp(-lp / sig))
lp <- predict(psm.weib, newdata = nd11, type = "lp")
p11 <- mySweibull(tseq, shape = 1 / sig, rate = exp(-lp / sig))

# pdf("AFT_Weibull_survival.pdf")
plot(tseq, p00, type = "l", lwd = 2, ylim = c(0, 1),
     xlab = "Years", ylab = "Survival")
lines(tseq, p01, type = "l", lwd = 2, col = 2)
lines(tseq, p10, type = "l", lwd = 2, col = 3)
lines(tseq, p11, type = "l", lwd = 2, col = 4)
legend("topright", c("No / no", "No / yes", "Yes / no", "Yes / yes"),
     lwd = 2, col = 1:4, bty = "n")
mtext("Hepato / edema", side = 3, adj = 1)
title(main = "Weibull AFT")
```



```
# dev.off()
```

Semi-parametric AFT models

It turns out you need to be quite careful with using time or $\log(\text{time})$ when calling different functions. Those that take the linear regression model as starting point need $\log(\text{time})$, because the outcome is $\log(\text{time})$, with the issue being that this outcome sometimes is right-censored.

```
tic("lss from {lss2}")
ls.lss <- lss(Surv(log(yrs), stat) ~ hepato + edema, data = pbc, mcs = 100)
```

```
##
## Converged. Criteria Satisfied: 0.001
```

```
toc()
```

```
## lss from {lss2}: 18.91 sec elapsed
```

```
ls.lss
```

```
##
## Call:
## lss(formula = Surv(log(yrs), stat) ~ hepato + edema, data = pbc,
```

```
##      mcsize = 100)
##
## Number of Observations:  312
## Number of Events:      144
## Number of Censored:    168
## Number of Iterations:   4
## Resampling Number:     100
##
## Gehan Estimator:
##      Estimate Std. Error  Z value    Pr(>|Z|)
## hepato -0.7807529  0.1528887 -5.106675 3.278776e-07
## edema  -1.4846471  0.2901092 -5.117546 3.095364e-07
##
## Least-Squares Estimator:
##      Estimate Std. Error  Z value    Pr(>|Z|)
## hepato -0.7795449  0.1505485 -5.178032 2.242382e-07
## edema  -1.4001970  0.2371112 -5.905233 3.521480e-09
```

The package aftgee uses time itself (not log(time)). Computation is much faster.

```
tic("aftgee from {aftgee}")
aft1 <- aftgee(Surv(yrs, stat) ~ hepato + edema, data = pbc)
toc()
```

```
## aftgee from {aftgee}: 0.21 sec elapsed
```

```
summary(aft1)
```

```
## Call:
## aftgee(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc)
##
## AFTGEE Estimator
##      Estimate StdErr z.value  p.value
## (Intercept)   2.692  0.127  21.238 < 2.2e-16 ***
## hepato        -0.780  0.157  -4.963 < 2.2e-16 ***
## edema         -1.400  0.209  -6.693 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The package aftsem again needs log(time) to be specified. It seems much faster than aftgee, but that is only when no SE's are computed.

```
tic("aftsem from {aftsem}, buckley")
aftsem1 <- aftsem(Surv(log(yrs), stat) ~ hepato + edema, data = pbc, method = "buckley")
toc()
```

```
## aftsem from {aftsem}, buckley: 0 sec elapsed
```

```
summary(aftsem1)
```

```
## AFT Semiparametric Model Summary
## =====
##
## Model Call: aftsem(formula = Surv(log(yrs), stat) ~ hepato + edema, data = pbc,
##   Model Call:      method = "buckley")
##
## Used parameter estimation method:
## [1] "buckley"
##
## Convergence Status: Not Converged
##
## Estimated Parameters:
##           [,1]
## [1,] -0.7789486
## [2,] -1.3997578
##
## Estimated Intercept:
## [1] 0
## Number of Iterations: 15
## Number of Observations: 312
## Percent of Censored Observations: 53.850000
##
## Initial Beta Estimate:
##      Zhepato      Zedema
## -0.2976519 -0.9589797
##
## Epsilon (Convergence Tolerance): 0.00001
## Maximum Iterations Allowed: 15
## Algorithm stopped after: 15 iterations
```

It turns out that 15 iterations is not enough for convergence (the same is true for the jin method coming up), but it is not really clear how to adjust this, so I am letting this go.

```
tic("aftsem from {aftsem}, jin")
aftsem2 <- aftsem(Surv(log(yrs), stat) ~ hepato + edema, data = pbc, method = "jin")
toc()
```

```
## aftsem from {aftsem}, jin: 0.15 sec elapsed
```

```
summary(aftsem2)
```

```
## AFT Semiparametric Model Summary
## =====
##
```

```

## Model Call: aftsem(formula = Surv(log(yrs), stat) ~ hepato + edema, data = pbc,
## Model Call:      method = "jin")
##
## Used parameter estimation method:
## [1] "jin"
##
## Convergence Status: Not Converged
##
## Estimated Parameters:
##      [,1]
## [1,] -0.779088
## [2,] -1.399838
##
## Number of Iterations: 15
## Number of Observations: 312
## Percent of Censored Observations: 53.850000
##
## Initial Beta Estimate:
##      [,1]
## [1,] -0.7807529
## [2,] -1.4846471
##
## Epsilon (Convergence Tolerance): 0.00001
## Maximum Iterations Allowed: 15
## Algorithm stopped after: 15 iterations
tic("aftsem from {aftsem}, jin, with 100 resamples")
aftsem2 <- aftsem(Surv(log(yrs), stat) ~ hepato + edema, data = pbc, method = "jin",
                  resample = 100)
toc()

## aftsem from {aftsem}, jin, with 100 resamples: 15.22 sec elapsed
summary(aftsem2)

## AFT Semiparametric Model Summary
## =====
##
## Model Call: aftsem(formula = Surv(log(yrs), stat) ~ hepato + edema, data = pbc,
## Model Call:      method = "jin", resample = 100)
##
## Used parameter estimation method:
## [1] "jin"
##
## Convergence Status: Not Converged
##

```

```

##           |Estimate| |Std. Error| |Z value| |Pr(>|Z|)|
## hepato    -0.779      0.150      -5.2    2.0e-07 ***
## edema     -1.400      0.222      -6.3    2.9e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## Number of Iterations: 15
## Number of Observations: 312
## Percent of Censored Observations: 53.850000

tic("aftsem from {aftsem}, gehan")
aftsem3 <- aftsem(Surv(log(yrs), stat) ~ hepato + edema, data = pbc, method = "gehan")
toc()

## aftsem from {aftsem}, gehan: 0.17 sec elapsed

summary(aftsem3)

## AFT Semiparametric Model Summary
## =====
##
## Model Call: aftsem(formula = Surv(log(yrs), stat) ~ hepato + edema, data = pbc,
## Model Call:      method = "gehan")
##
## Used parameter estimation method:
## [1] "gehan"
##
## Convergence Status: Converged
##
## Estimated Parameters:
##           [,1]
## [1,] -0.7807529
## [2,] -1.4846471
##
## Number of Observations: 312
## Percent of Censored Observations: 53.850000
##
## Initial Beta Estimate:
## Not available

tic("aftsem from {aftsem}, gehan, with 100 resamples")
aftsem3 <- aftsem(Surv(log(yrs), stat) ~ hepato + edema, data = pbc, method = "gehan",
                  resample = 100)
toc()

## aftsem from {aftsem}, gehan, with 100 resamples: 15.3 sec elapsed

```

```
summary(aftsem3)
```

```
## AFT Semiparametric Model Summary
## =====
##
## Model Call: aftsem(formula = Surv(log(yrs), stat) ~ hepato + edema, data = pbc,
##   Model Call:      method = "gehan", resample = 100)
##
## Used parameter estimation method:
## [1] "gehan"
##
## Convergence Status: Converged
##
##      |Estimate| |Std. Error| |Z value| |Pr(>|Z|)|
## hepato    -0.781      0.155    -5.03    4.8e-07 ***
## edema     -1.485      0.259    -5.74    9.4e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## Number of Observations: 312
## Percent of Censored Observations: 53.850000
```

```
tic("aftsem from {aftsem}, gehan-heller")
```

```
aftsem4 <- aftsem(Surv(log(yrs), stat) ~ hepato + edema, data = pbc, method = "gehan-heller")
```

```
toc()
```

```
## aftsem from {aftsem}, gehan-heller: 0.98 sec elapsed
```

```
summary(aftsem4)
```

```
## AFT Semiparametric Model Summary
## =====
##
## Model Call: aftsem(formula = Surv(log(yrs), stat) ~ hepato + edema, data = pbc,
##   Model Call:      method = "gehan-heller")
##
## Used parameter estimation method:
## [1] "gehan-heller"
##
## Convergence Status: Converged
##
## Estimated Parameters:
##      [,1]
## [1,] -0.8945909
## [2,] -1.4988955
##
## Number of Observations: 312
```



```
## Percent of Censored Observations: 53.850000
##
## Initial Beta Estimate:
##      Zhepato      Zedema
## -0.2976519 -0.9589797

tic("aftsem from {aftsem}, gehan-poly")
aftsem5 <- aftsem(Surv(log(yrs), stat) ~ hepato + edema, data = pbc, method = "gehan-poly")
toc()
```

```
## aftsem from {aftsem}, gehan-poly: 0.21 sec elapsed
```

```
summary(aftsem5)
```

```
## AFT Semiparametric Model Summary
## =====
##
## Model Call: aftsem(formula = Surv(log(yrs), stat) ~ hepato + edema, data = pbc,
## Model Call:      method = "gehan-poly")
##
## Used parameter estimation method:
## [1] "gehan-poly"
##
## Convergence Status: Converged
##
## Estimated Parameters:
##           [,1]
## [1,] -0.7807396
## [2,] -1.4846315
##
## Number of Observations: 312
## Percent of Censored Observations: 53.850000
##
## Initial Beta Estimate:
##      Zhepato      Zedema
## -0.2976519 -0.9589797
```

Finally, we have the Buckley-James estimator, implemented in the rms package. This package uses time, not log(time).

```
tic("bj from {rms}")
ls.bj <- bj(Surv(yrs, stat) ~ hepato + edema, data = pbc)
toc()
```

```
## bj from {rms}: 0.01 sec elapsed
```

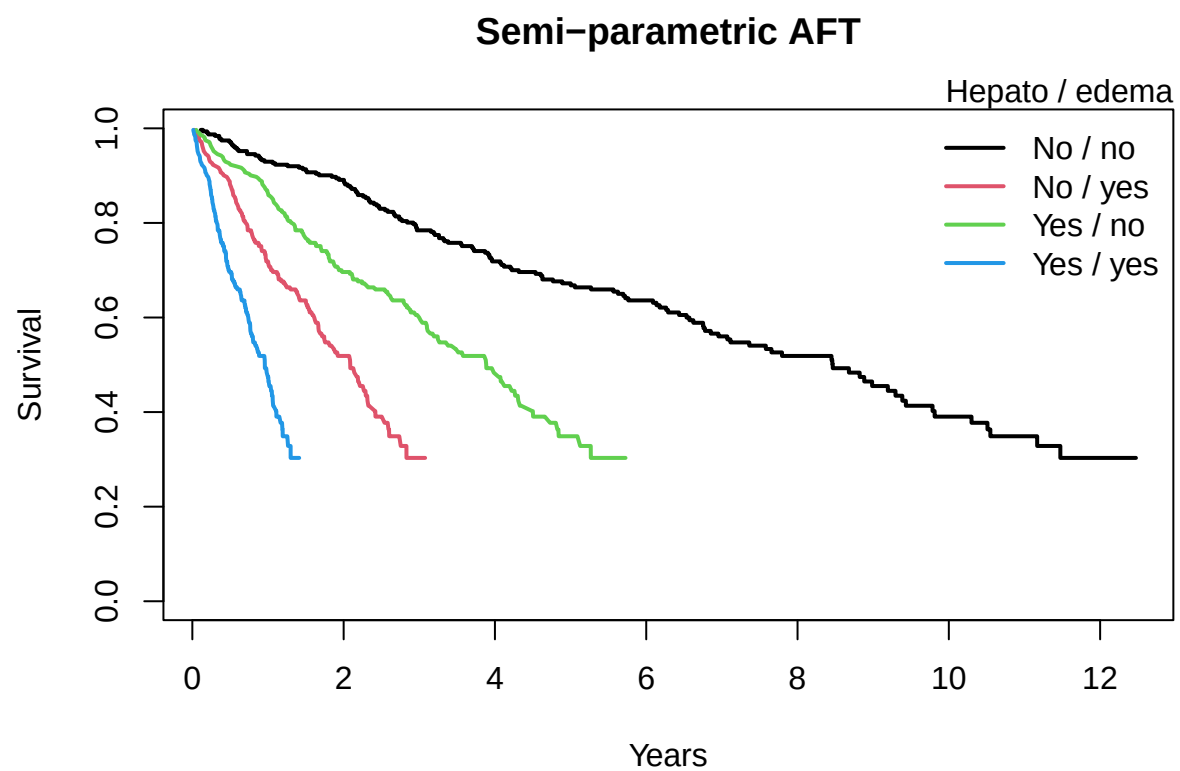
```
ls.bj
```

```
## Buckley-James Censored Data Regression
##
## bj(formula = Surv(yrs, stat) ~ hepato + edema, data = pbc)
##
##                               Discrimination
##                               Indexes
## Obs      312      Regression d.f.2      g      0.706
## Events 144      sigma0.9103      gr      2.026
##                               d.f.      141
##
##      Coef      S.E.      Wald Z Pr(>|Z|)
## Intercept  2.6906 0.1400 19.22 <0.0001
## hepato     -0.7786 0.1668 -4.67 <0.0001
## edema      -1.3997 0.1743 -8.03 <0.0001
```

Here is an attempt to obtain model-based survival curves from the Buckley-James estimator (obtained from `ls.bj`).

```
scaled_yrs <- pbc$yrs
S00 <- survfit(Surv(scaled_yrs, pbc$stat) ~ 1)
scaled_yrs <- pbc$yrs * exp( ls.bj$coef[3] )
S01 <- survfit(Surv(scaled_yrs, pbc$stat) ~ 1)
scaled_yrs <- pbc$yrs * exp( ls.bj$coef[2] )
S10 <- survfit(Surv(scaled_yrs, pbc$stat) ~ 1)
scaled_yrs <- pbc$yrs * exp( ls.bj$coef[2] + ls.bj$coef[3])
S11 <- survfit(Surv(scaled_yrs, pbc$stat) ~ 1)

# pdf("AFT_semipar_survival.pdf")
plot(S00$time, S00$surv, type = "s", lwd = 2, ylim = c(0, 1),
     xlab = "Years", ylab = "Survival")
lines(S01$time, S01$surv, col = 2, lwd = 2)
lines(S10$time, S10$surv, col = 3, lwd = 2)
lines(S11$time, S11$surv, col = 4, lwd = 2)
legend("topright", c("No / no", "No / yes", "Yes / no", "Yes / yes"),
     lwd = 2, col = 1:4, bty = "n")
mtext("Hepato / edema", side = 3, adj = 1)
title(main = "Semi-parametric AFT")
```



```
# dev.off()
```